Chapter 9

The thoracic aorta

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Surgical options

A variety of operative approaches can be used in thoracic aortic disease. See Table 9.1.

Table 9.1 Surgical options

Technique	Indication	Summary
Fenestration	Persistent organ ischemia after surgical correction of the proximal tear.	Creating holes in the dissection flap so that blood can perfuse organs being supplied by both true and false lumens. Can be done open or via percutaneous approach.
AVR and interposition graft	Abnormal aortic valve and aneurysmal or dissected ascending aorta.	AVR carried out as described on 📖 pp378–382. Then interposition graft placed (see 📖 p444)
Wolfe procedure	Aneurysm or dissection involving non-coronary sinus, but not the left or right	Similar to interposition graft, but with proximal scallop to recreate non-coronary sinus
Bentall technique	Abnormal aortic valve and ascending aneurysm/ dissection/endocarditis involving annulus and/or sinuses	See 📖 p390. In the original 1968 description the coronary ostia were sutured directly to holes created in the tube graft without making buttons, and the conduit was not composite
Modified Bentall	As above	Same as the Bentall but with composite valved graft, and with the coronaries excised with a cuff to create 'buttons' and mobilized proximally (📖 p390)
Cabrol graft	Used where coronary buttons cannot be mobilised adequately e.g., calcification, previous root replacement	Valved conduit implanted in same way as Bentall, but 8mm Dacron graft anastomosed end to end to left coronary buton, then either to aorta (hemi-Cabrol) or end to side to right coronaru buttom (full Cabrol) and then to aorta (📖 p445)
Elephant trunk	Arch aneurysm/ dissection where descending aorta also needs replacement. 📖 p448	Two-stage procedure where distal end of Dacron graft is left inside descending aorta after arch replacement, for later replacement of descending aorta. (📖 pp448–451)
Homograft	Similar indications to Bentall, valuable where tissue very friable e.g., endocarditis	Described on 📖 p386

Table 9.1 (Cont'd)

Technique	Indication	Summary
Trifurcation graft	Alternative to "island technique" (Fig. 9.7b), for arch replacement (📖 pp448–449)	
Glue aortoplasty	Obliterate false lumen in dissection, most commonly to treat extension into aortic sinuses in conjunction with interposition grafts.	Biological glues used

Endoluminal stenting is described on 📖 p431.

Pathology of aortic dissection

Definition

Blood leaves the aortic lumen via an intimal tear, separating the inner from the outer layers of the media creating a false lumen.

Classifications of aortic dissection

- Acute dissections are those presenting within 14 days.
- Chronic dissections present after 2 months.
- Subacute dissections present between 14 days and 2 months.

Stanford (Fig. 9.1)

Type A: any case in which the ascending aorta is involved.
Type B: any case not involving the ascending aorta.

Debakey (Fig. 9.1)

Type I: tear in the ascending aorta, dissection extending to arch.
Type II: tear in ascending aorta, dissection limited to ascending aorta.
Type IIIa: tear in descending aorta, dissection limited to descending.
Type IIIb: tear in descending aorta, dissection may extend to arch and ascending aorta.

Emergency surgery is indicated for acute Stanford type A and DeBakey type I and II dissections.

Risk factors

The pathogenesis of aortic dissection is not well understood. There are several theories that might explain the etiology of the intimal tear that results in dissection. An early theory was cystic medial degeneration where the media is abnormal, but as this abnormality has been found in only a minority of patients with acute aortic dissection, this theory has lost ground. Later theories identified atherosclerotic ulceration penetrating to the media as the source of the intimal tear, but again these are uncommon at the site of the intimal tear. Finally, intramural hematoma resulting from bleeding of the vasa vasorum into the media is thought to account for sufficient stress to cause an intimal tear. Risk factors include:

- Age (up to 90% of patients are >60 years of age).
- Hypertension (chronic primary hypertension, and Sheehan and Cushing syndromes, pheochromocytoma, polycystic kidney disease).
- High cardiac output states, e.g. pregnancy.
- Connective tissue disorders (Ehlers–Danlos syndrome, Marfan syndrome, Turner syndrome, 📖 p429).
- Atherosclerosis is a risk factor only in pre-existing aneurysms.
- Bicuspid aortic valve, cystic medial degeneration, aortitis.
- Iatrogenic (catheterization, cannulation, aortic cross-clamping, AVR, proximal aorto-coronary anastomosis, IABP).
- Competitive weightlifting.
- Trauma.

Clinical features

Dissection usually results in sudden, severe, 'tearing' pain radiating from chest through to the back, and sometimes neck and arms. About 30–40% die immediately. A minority of patients present with asymptomatic, usually chronic, dissections picked as incidental findings on CT. Complications of aortic dissection include:

- Arch vessel occlusion causes stroke in 5% of type A dissections.
- Myocardial ischemia occurs in up to 20% of type A dissections.
- Hemopericardium occurs in about a 1/4 of type A dissections.
- Severe AI occurs in about 1/3 of type A dissections.
- Occlusion of vessels may lead to upper and lower limb, mesenteric, and renal ischemia: peripheral vascular examination is abnormal in 20%.
- Paraplegia develops in 2–5% of type B dissections as a result of spinal artery occlusion.
- Rarely acute aortic dissection may cause SVC syndrome, hemoptysis, airway compression, and Horner's syndrome.

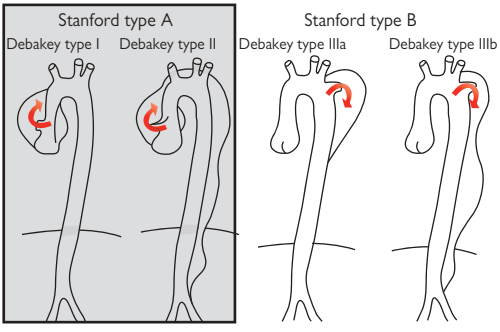


Fig. 9.1 DeBakey and Stanford classifications of aortic dissection.

Pathology of aortic aneurysms

Key facts

An aneurysm is defined as a permanent localized dilatation of an artery by at least 50% compared to normal. The dimensions of the normal aorta are (uncorrected for body mass index):

- Annulus: 2.8–3.2 cm (2.1cm/m^2 adjusted for BSA).
- Sinus of Valsalva: 3.4 ± 0.3 in men, 3.0 ± 0.3 in women.
- Sinotubular junction: 2.4–3.0cm (1.9cm/m^2, or roughly 0.9x annulus).
- Mid ascending aorta: 2.6–3.4cm.
- Mid arch 2.4–3.1 cm.
- Mid descending aorta 2.8cm in men, 2.6cm in women.
- At the celiac axis 2.3cm in men, 2.0cm in women.
- Infrarenal aorta 1.9cm in men, 1.5cm in women.

Classifications of aortic aneurysm

- A *true aneurysm* involves all layers of the arterial wall.
- A *false or pseudoaneurysm* usually has an outer wall consisting of adventia, organized thrombus, and surrounding structures.
- A *mycotic aneurysm* is any aneurysm caused as a result of infection.
- Aneurysms can be further classified by location: ascending, arch, descending, abdominal using the *Crawford classification* (Fig. 9.2).
- Aneurysms are classified according to shape (*fusiform or saccular*).
- Aneurysms may be classified according to complicating pathology: (*ruptured, dissecting*).

Risk factors

Aortic aneurysm formation is usually due to loss of elasticity and tensile strength in the medial layer of the aorta, which is normally composed of elastin, collagen, smooth muscle cells which degrade and synthesize elastin, and ground matrix. Elastin content is highest in the ascending aorta and the media becomes thinner distally. In typical aortic aneurysms there is fragmentation of elastin fibres within the media and loss of smooth muscle cells. This is called *cystic medial degeneration*. The ascending aorta expands and recoils with systole and diastole respectively, maximizing forward flow. When elasticity is lost, the aorta begins to dilate. The aorta dilates normally with age. Risk factors for aneurysm formation include:

- Age (up to 90% of patients are >60 years of age).
- Hypertension, atherosclerosis, smoking.
- Connective tissue disorders (Ehlers–Danlos syndrome, Marfan syndrome, Turner syndrome, Loeys–Dietz syndrome 📖 p429).
- Cystic medial degeneration.
- Aortitis (Takayasu's arteritis most commonly involves the arch).
- Aortic stenosis, bicuspid aortic valve: two mechanisms—(1) post-stenotic dilatation, (2) increased medial degeneration in bicuspid AV so expansion is faster and risk of aortic aneurysm and dissection 10x that

of normal population (1:1000 vs. 1:10 000). Aortic aneurysm is still rare in patients with bicuspid valve, but they represent a high proportion of aneurysm patients.
- Syphilis (characteristically ascending aneurysms).
- Bacterial infection (most commonly *Staphylococcus aureus, Staphylococcus epidermidis, Salmonella, Streptococcus*).
- Trauma is a rare cause of true aneurysms, but iatrogenic trauma (surgery or intervention) is a common cause of pseudoaneurysms.

Connective tissue disorders
- *Marfan syndrome* is an autosomal dominant connective tissue disorder with an incidence of 1:10 000 which causes a fault in the fibrillin gene of chromosome 15: abnormal *elastin* fibres are prone to disruption and cystic medial degeneration occurs at an early age; commonest cause of death is aortic dissection.
- *Ehlers–Danlos syndrome* is an inherited disorder of connective tissue with several subtypes of which type IV is associated with cardiovascular malformations due to defect in *collagen* formation: rupture of mesenteric vessels more commonly than thoracic aneurysm formation.
- Familial aneurysms that are not due to Marfan or Ehlers–Danlos syndromes are aneurysms occurring in families with a strong history of aortic aneurysm and dissection, and may be linked to the 5q locus.

Clinical features
- About 1/3 of patients are asymptomatic: the aneurysm is an incidental finding on CXR, or TTE or cardiac catheterization.
- About half of patients present with anterior chest pain.
- Acute onset pain suggests dissection or impending rupture.
- Hoarseness from arch aneurysms stretching recurrent laryngeal nerve.
- Examination is frequently unremarkable but 10–20% of patients have associated abdominal aneurysms.

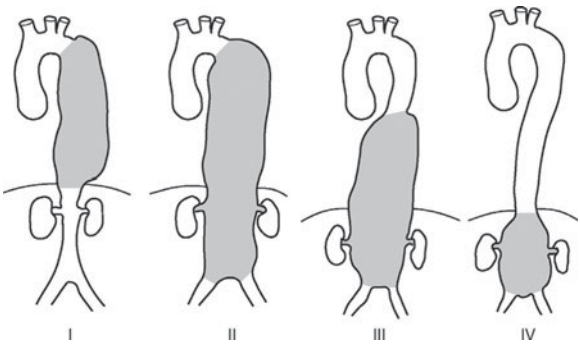


Fig. 9.2 Crawford classification of thoracoabdominal aortic aneurysms.

Diagnosis of type A dissection

Imaging

- The primary role of the diagnostic modalities is to differentiate between a type A dissection (an indication for emergency surgery) and a type B dissection. A secondary role is assessing the extent of dissection which dictates surgical approach.
- As a broad rule, centers planning to refer suspected acute type A dissections should use the imaging modality in which they have most expertise that can be done with minimum delay: an expertly performed TEE is more sensitive and specific, for example, than an inexpertly performed and interpreted CT angiogram (CTA); neither should delay referral.

Table 9.2 Imaging modalities used in diagnosis of type A dissection

Imaging modality	Sensitivity	Specificity	Comments
TEE + Doppler	96–100%	86–100%	Widely available, familiar. Dynamic images generated. Shows coronary ostia, LV function, AI, pericardium, pleura
CT angiography	90–100%	90–100%	Quick, readily available, familiar Shows pleural + pericardial space Shows head and neck vessels IV contrast may cause renal failure
MRI	98–100%	98–100%	Gold standard Detailed dynamic images of aorta, pericardium, pleura, LV, AV, flow in branch vessels and false lumen Not easily available or accessible Contraindications (📖 p440)
Aortography	80–90%	88–95%	Until recently gold standard Facilities and expertise not as widely available Arterial access often difficult Can precipitate rupture Shows perfusion of true and false lumens, flap, coronary arteries, AI

Investigations

- In the hemodynamically unstable patient where there is a high index of suspicion (e.g., in the patient with crushing chest pain already known to have an aneurysm) it is reasonable to transfer directly to the operating room, anaesthetize and perform a TEE. If the TEE is positive, lines are placed so that it is possible to proceed directly to surgery: if the TEE is negative, invasive monitoring will allow the patient to undergo additional investigation.
- In more stable patients a CTA can be obtained quickly, with a plan to transfer the patient directly to the operating room, without delaying to insert monitoring lines or treat pressure if the CTA is positive.
- Blood tests are usually normal: raised creatinine and LFTs sometimes indicate chronic hypoperfusion states, and raised troponin suggests involvement of coronary ostia.
- EKG should be inspected for signs of ischemia, but in 30% of patients in whom the coronary ostia are involved EKG changes are non-specific.
- CXR may show a widened mediastinum, cardiomegaly secondary to pericardial effusion, and signs of pleural effusion (usually left).
- Coronary angiography is difficult and hazardous, and the same information can be obtained by CTA: most patients with type A dissections go to surgery without cardiac catheterization.

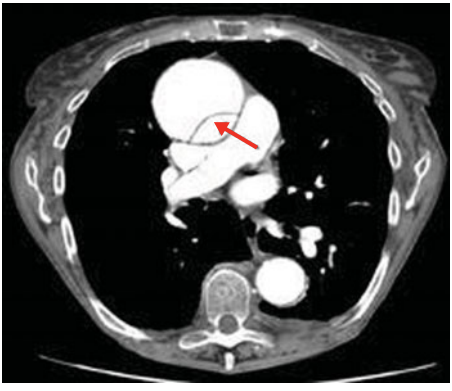


Fig. 9.3 CT scan of a type A dissection. A Type B dissection is shown on 📖 p17 (Fig. 1.4 and b).

Management of type A dissection

Natural history

- 30% of patients die immediately, 50% die within 48h, medical management has a 60% in-hospital mortality. Operative mortality is 20–30%. 1-year survival is about 66%, 5-year survival is 45–55%.
- Mode of death in acute presentation may be ***rupture, cardiac tamponade, AMI, stroke, cardiogenic shock from acute severe AI.***

Indications for surgery

- The aim of surgery is to treat/prevent the causes of death.
- An acute type A dissection is an indication for emergency surgery.
- A subacute type A dissection is an indication for urgent surgery.

Contraindications to surgery

- Advanced age (>85 years) or frailty are relative contraindications.
- Incurable malignancy or other terminal illness.
- Chronic dementia or irreversible, profound brain injury.
- AMI, coma, stroke, paraplegia, renal failure, acute limb and visceral ischemia increase risks of surgery, but may improve following repair of the dissection, so are not contraindications to surgery.

Planning surgery for type A dissection

The main variables to think about are: (1) the ***extent of resection*** (root or arch replacement required?), (2) ***cannulation*** strategy, (3) need for ***circulatory arrest***, (4) can the aorta be clamped (dictates ***sequence*** of distal vs. proximal anastomosis as well as need for DHCA), (5) ***myocardial protection*** strategy, (6) ***venting***. The impact of each on the conduct of surgery is outlined on 📖 pp433–435. There are many possible scenarios:

- The arch and root are normal: a simple interposition graft using aortocaval cannulation, direct ostial ± retrograde cardioplegia, no DHCA.
- Distal aorta aneurysmal: interposition graft and hemiarch replacement with DHCA using axillary cannulation, ostial ± retrograde cardioplegia.
- Root aneurysmal: root replacement with or without DHCA.
- Arch involved: DHCA ± selective or retrograde cerebral perfusion.

Monitoring lines

At least two arterial lines are helpful as dissection or cannulation of arch and distal vessels may lead to misleading arterial pressure measurements.

- A right radial line is useful in right axillary cannulation to detect hyper- (or hypo-) perfusion of the right arm if a side-graft cannulation technique is used, but pressures transduced from it during bypass may bear little correlation with systemic pressures, especially in direct right axillary artery cannulation (📖 p120).
- Left axillary or femoral lines give more accurate central arterial pressure than a radial line which is usually damped by end of case.

Prepping and draping

- Drape for a full sternotomy, but include the lower neck in the field in case the incision has to be extended for arch involvement, and the whole infraclavicular area to allow axillary cannulation.

- Shave and prep both groins for possible cannulation.
- Prep and expose thighs in case saphenous vein is required.

Incision

- Type A dissections are repaired via a median sternotomy: may need to extend up into neck, so always prep and expose this area as well.

Cannulation

Very large aneurysms may make access to RA difficult, and may preclude safe aortic cannulation. The main options are:

- Central cannulation with venous return via a two-stage cannula in the RA and arterial return to the distal ascending aorta or arch (fastest option in an emergency, can use TEE to identify true lumen).
- Fem–caval cannulation, with the femoral arterial cannula placed either before or after opening the chest (if dissection extends to the femoral vessels it is important to cannulate the lumen in which blood is flowing, irrespective of whether it is the true or false lumen). Next fastest.
- Fem–fem cannulation: may be favoured if rupture suspected, as patient can be cooled in readiness for DHCA without opening pericardium, which may precipitate acute rupture.
- Axillo-caval cannulation: axillary cannulation usually before sternotomy, slowest of all options in an emergency, but less risk of stroke or cannulating false lumen, and facilitates selective cerebral perfusion (📖 p125).
- Use bicaval cannulation if retrograde cerebral perfusion is planned.

Myocardial protection

- Depending on the nature of the ascending aorta it may be preferable not to clamp it: bypass is commenced and the patient cooled, the heart fibrillates around 32°C, and retrograde cardioplegia is given at regular intervals to supplement cardioprotection from profound systemic hypothermia. If there is AI the LV must be vented to prevent distention.
- Antegrade can be given directly via coronary ostia after aortotomy.

Venting

- The LV via the right superior pulmonary vein (usually much less difficult to access once bypass commenced) or LV apex.
- Pump sucker placed in the LV via the aortic valve.

Cerebral protection

- Cool to 18°C (usually safe to assume DHCA will be needed).
- DHCA mandatory (📖 p125) if the arch is involved, or if the distal repair may only be performed with the cross-clamp off.
- Cerebral perfusion during DHCA may be none, retrograde, or from selective perfusion (📖 p125).

Overview of initial set-up

- Ensure the patient is monitored with two or more arterial lines.
- Prep and drape the patient, cannulate right axillary or femoral artery.
- Open the chest, give full-dose heparin, and then place purse-strings.
- Cannulate RA, go on bypass, start cooling, separate aorta from the PA.
- Either cross-clamp the distal ascending aorta or open the ascending aorta once the required systemic temperature is reached.
- Give cardioplegia directly into the coronary ostia or retrograde.

Repair of type A dissection

Key facts

- In the acute setting it is usually safest to focus on treating the immediately life-threatening complication of ascending aortic rupture: aortic arch resection is limited to the proximal arch, and valve involvement addressed by resuspension of the commissures rather than attempting root replacement. If the dissection extends into the coronary sinuses bioglue and a careful proximal suture line on the interposition graft should prevent progression, as the coronary ostia are usually spared.
- If it is safe to clamp the ascending aorta (avoid in large aneurysms where rupture is a risk); the proximal anastomosis can be completed while the patient is cooled.
- The distal anastomoses can be performed with the cross-clamp on (no need for circulatory arrest) or off (deep hypothermic circulatory arrest (DHCA) required), depending on how far distally the dissection extends, and whether the aorta can be safely clamped.
- Cannulation options are discussed on 📖 p433: aorto-caval is fastest in emergencies but may be contraindicated: femoral-caval is next fastest, followed by axillary cannulation which takes more time, but minimizes risk of cannulating false lumen and stroke, and permits selective cerebral perfusion during arch replacement without moving cannula.

Interposition graft

This technique is preferred for DeBakey type II dissections, not involving the aortic root. It supersedes the older graft inclusion technique, where aortic tissue was preserved and wrapped around the interposition graft.

- Set-up is described on 📖 pp432–433.
- Bypass is established via aorto-caval cannulation.
- The perfusionist begins to cool the patient to 28°C.
- The aortic clamp is placed just proximal to the innominate artery.
- Give cardioplegia anterogradely via a Medicut cannula into the root, or retrogradely via a coronary sinus cannula, to arrest the heart,
- Open the aorta transversely and use a pump sucker to vent the heart.
- Inspect the aortic lumen and the aortic valve: if the dissection extends past the sinotubular junction this will need reconstruction, if intimal tear extends distal to the cross-clamp circulatory arrest is required to inspect and repair the distal ascending aorta, in which case continue cooling to 18°C.
- Give cardioplegia directly into the coronary ostia and/or retrograde.
- Transect the aorta 5–10mm distal to the sinotubular junction (Fig. 9.4).
- If the dissection extends to the sinotubular junction reconstruct the aorta using bioglue to reapproximate the intima and media, and secure the whole thing with 4/0 Prolene running stitches anchoring a strip of PTFE felt all the way around the proximal aorta.
- Size the proximal aorta: size 26–28 usually appropriate.
- Suture the proximal aorta and felt to a Dacron graft with double-ended 4/0 Prolene RB needle running suture, starting forehand inside

to out, aorta to graft, a few bites further away from 6 o'clock (Fig. 9.4), running towards you until you need to take bites in two, then taking other arm and run it up the wall furthest away from you in two bites, then across the top towards you forehand in single bites. It's usually worth tightening the whole anastomosis with nerve hooks before tying.

- Transect the distal aorta and bevel the Dacron tube to match size.
- If the distal anastomosis is to be carried out without the cross-clamp, stop bypass when the temperature is 18°C, clamp lines, stop vents, and remove the cross-clamp: this anastomosis should take <20min.
- Suture the distal cuff of ascending aorta to the distal graft, with felt strip, in the same way as the proximal aorta (starting forehand inside to out, graft to felt to aorta, a few bites further away from 6 o'clock (Fig. 9.4), running towards you until you need to take bites in two, then take other arm of suture and run it up the wall furthest away from you in two bites, then forehand across the top. If size mismatch cheat on the top, never the bottom where leaks are awkward to fix.
- As the distal anastomosis is completed the graft is allowed to fill retrograde, remove the arterial line clamp and ask the perfusionist to come on at 500mL of flow, shake the graft so it de-airs fully and suction debris, come up further on flows and tie, then come fully up, unclamp venous line and vents, and resume bypass.
- Place a vent in the graft, and carry out standard de-airing (📖 p114).

Hemostasis

- Look for any 'spurters' and fix with 4/0 pledgetted Prolene with the flows down. For everything else pack and reassess ***after*** protamine has been given *and* any coagulopathy (usually platelet dysfunction) ***completely*** corrected: using Prolene to treat non-surgical bleeding begets more bleeding and coagulopathy. ***Cabrol patch*** is described on 📖 p445.

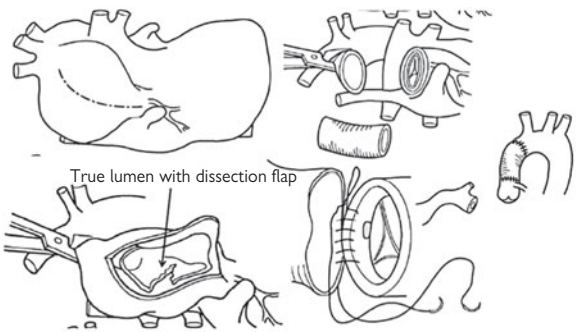


Fig. 9.4 Technique of performing an interposition graft for a type A dissection. Although a longitudinal aortotomy is shown, a transverse aortotomy is more commonly performed.

Management of type B dissection

Natural history

- Type B dissections have an initial hospital mortality of about 10%. About 2/3 of remaining patients have no complications requiring operation. 50% of unoperated patients, however, die within 3–5 years, of rupture or complications related to end-organ ischemia.
- Operative mortality is 5–15%, with a 5–30% risk of paraplegia depending on the presentation. After operation 1-year survival is about 65%, 5-year survival is 50%, 10-year survival 30%.

Indications for surgery

Medical treatment is primarily aimed at maintaining low arterial BP and reducing force of ventricular ejection. Labetolol (250mg in 250mL 5% dextrose at a rate of 2–8mg/min) is the best agent for this in hospital. Complications requiring emergency surgery include:

- Aortic rupture, periaortic or mediastinal hematoma, hemothorax.
- Dissection in pre-existing aneurysm.
- Rapid increase in aortic diameter.
- Ischemia of limbs and viscera.
- Intractable pain despite optimal medical management, progression.
- Marfan syndrome.

Contraindications to surgery

- Advanced age (>80 years) or frailty are relative contraindications.
- Incurable malignancy or other terminal illness.
- Chronic dementia.
- Severe pulmonary disease favours stenting over surgery.
- Irreversible, profound brain injury.
- New stroke, paraplegia, renal failure, acute limb and visceral ischemia increase the risks of surgery, but as they may improve following repair of the dissection, they are not contraindications to surgery.

Complications of surgery

- Death 10–15% (up to 50% in elderly patients with multiple comorbidity).
- Stroke 5%, paraplegia 5–10% (acute dissection ~30%, chronic dissection ~10%).

Spinal cord protection

Spinal cord perfusion depends on an adequate difference between MAPs and intracranial pressure (dictated by CVP and CSF pressure). This may be optimized perioperatively by focusing on several areas:

Arterial supply

The blood supply to the spinal cord is from vertebral arteries, and from radicular arteries which are segmental branches of the thoracic and abdominal aorta. Each vertebral artery gives rise to a posterior and an anterior spinal artery: the two anterior spinal arteries fuse to form a single midline vessel which travels along the median fissure of the spinal cord. Below the neck all three spinal arteries unite to form a continuous

series of anastomoses with posterior intercostal (radicular) arteries. In its lower segments the spinal cord depends more on radicular arteries. Some surgeons believe a consistent and large radicular artery (artery of Adamkiewicz) present at T12–L2 provides the arterial supply for the lower 2/3 of the cord, but it has not been demonstrated consistently.

Surgical technique

There are several approaches. *Preventing hypotension* is key to all.

- *Reimplant large radicular arteries* below particularly below T10 (disadvantages include longer ischemic time and risk of bleeding).
- *Monitor somatosensory evoked potentials (SSEPs) and motor evoked potentials (MEPs)* which give a better indication of anterior cord function, clamping intercostals vessels sequentially but only dividing them if no change in SSEP is seen after 10min of clamping.
- *Minimize ischemic time by perfusion of vessels downstream* of the distal graft anastomosis rather than 'clamp and go' technique.

Cooling and hypothermia

Systemic hypothermia offers some protection to cord ischemia primarily by reducing cord temperature during the ischemic period: systemic hypothermia is more widely used than surface and intrathecal cooling.

Cerebrospinal fluid withdrawal

Intra- and postoperative withdrawal of cerebrospinal fluid to maintain CSF pressure 10–20mmHg is aims to increase cord perfusion by increasing the difference between intracranial and arterial pressure.

Pharmacological methods

The most important pharmacological adjuncts are vasopressors, given to *maintain MAPs 90–100mmHg* postoperatively: late onset paraplegia is associated with ↓BP. Corticosteroids, sodium thiopental, prostaglandin E, and O_2 free radical scavengers are adjuncts of unproven importance.

Endoluminal stenting

A proportion of patients with acute or chronic type B dissections are candidates for endoluminal stenting.

Indications

For the stent to remain *in situ* there needs to be a neck of 2cm of normal aorta both proximally and distally. The precise indications for endoluminal stenting still a matter of debate, but most patients, particularly older patients with multiple comorbidity, should be considered for stenting for the following purposes:

- Stenting of true lumen to seal entry tear.
- Stenting of static obstruction of a branch artery.
- Balloon fenestration of dissection flap and stenting of true lumen for visceral or lower limb hypoperfusion.
- Stenting of true lumen to enlarge compressed true lumen.

Complications of endoluminal stenting

- Hospital mortality (MI, aneurysm rupture) 5–10%.
- Paraplegia 5–10% (depends on morphology), stroke 2–5%.
- Stent migration 2–5%, organ ischemia 2–5%.

Repair of type B dissection

Key facts

- The aim of repair is to resect as much of the involved aorta as possible while minimizing risk of spinal cord and end organ ischemia.
- The tear is usually at the origin of the left subclavian artery. The aorta containing the intimal tear must be resected, but it is frequently not possible to obliterate the entire false lumen.
- Repair can be done off bypass, on bypass, with or without deep hypothermic circulatory arrest, or partial left heart bypass depending on the extent of the dissection, with or without renal artery perfusion:
  - ***Full bypass*** allows control of BP, and option of ***DHCA*** which improves ability to control proximal portion (especially if arch involved), eliminates bleeding collaterals, as well as allowing BP control and lower body to be perfused.
  - ***Partial left heart bypass*** (📖 p125) also allows BP control and distal perfusion, but without coagulopathy associated with full heparinization and use of oxygenator. Arterial return is to the femoral artery, with venous drainage from the left inferior pulmonary vein, or left atrial appendage.
  - ***'Clamp and go'*** avoids the coagulopathy and inflammatory picture associated with bypass, but relies on a fast anastomosis as the lower body warm ischemic time is longest with this technique, and risks upper body hypertension once the descending aorta is clamped.
- Spinal cord protection strategies aim to minimize paraplegia risk (📖 p437).

Set-up

- Ensure that the patient has external defibrillator paddles, right and left radial arterial lines (the left subclavian may be snared or clamped), and that the groins are shaved.
- A double-lumen ET tube facilitates left lung decompression.
- Prep and drape the patient in the right lateral position, with the pelvis as horizontal as possible to allow access to both groins.
- Make a left posterolateral thoracotomy incision in the 5th intercostal space, and divide the costotransverse ligaments of the 5th and 6th rib to permit maximum exposure without requiring rib fracture (📖 p39).
- A thoracoabdominal incision (Fig. 9.5) is required if the abdominal aorta is involved: the abdominal exposure is normally retroperitoneal but peritoneum can be opened if there is a need to view viscera, e.g., concern about ischemia.
- Once the thoracic aorta is exposed secure the left subclavian artery, and preserve the ***left vagus and recurrent laryngeal nerves***.
- The thoracic aorta is freed from the left common carotid artery as far as the distal extent of the resection, clipping and dividing the spinal arteries in the segment to be resected.
- If left heart bypass is to be used, open the pericardium and place a 4/0 Prolene purse-string in the left inferior pulmonary vein and cannulate with a 14F cannula, with arterial return to either the distal descending aorta, or the femoral artery providing flow rates of 1–2L/min.
- If total cardiopulmonary bypass is required, use fem–fem cannulation.

Repair

- Once fully heparinized go on bypass and place vascular clamps on the aorta proximal and distal to the area of resection.
- Maintain right radial artery pressures at 100–130mmHg and femoral artery pressures at 60–70mmHg.
- Open the aorta longitudinally and ligate bleeding intercostals arteries: this can be done sequentially while monitoring SSEPs and MEPs (📖 p437).
- Transect the aorta at least 1cm distal to the origin of the left subclavian artery.
- For proximal anastomosis use pledgetted 4/0 Prolene (Fig. 9.5): in a stage II elephant trunk (see 📖 p449 for stage I), the proximal anastomosis is to the distal end of the Dacron graft left at the previous operation, which is located and orientated under a brief period of circulatory arrest with clamp off.
- When this anastomosis is complete, reposition the proximal clamp on the graft which is allowed to fill with blood, so that the anastomosis can be inspected (Fig. 9.5).
- If the proximal anastomosis is satisfactory, transect the distal aorta, cut the graft to fit and perform the anastomosis with 3/0 Prolene, Teflon strips, and biological glue as indicated.
- Depending on the level and extent of resection, renal and celiac arteries may be reimplanted into the Dacron graft at this stage.
- Wean bypass, reverse heparin, and secure hemostasis.

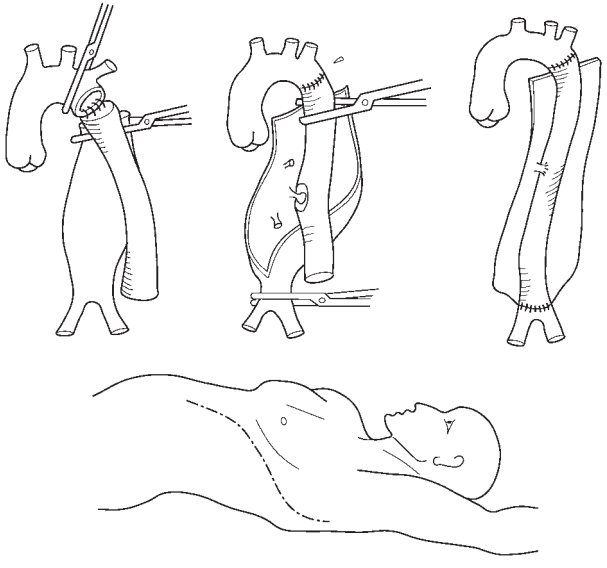


Fig. 9.5 Technique of repair of descending aorta.

Diagnosis of aortic aneurysm

Key facts

- The role of imaging in chronic aneurysms is to establish and monitor the size of the aneurysm, and define the anatomy and comorbidity.
- CT is the most widely used modality, but MRI is a useful adjunct.
- Additional assessment includes echocardiography and coronary angiography.

Imaging

CT

CT scanners are widely available and the images familiar to all clinicians. CT provides accurate and reproducible information about size, anatomy, and extent of the disease, so it is used for 6-monthly assessment of chronic aneurysms: these must be *standardized assessments* as estimates of aneurysm size can vary widely between scans, and within a single test depending on where the measurement was taken and how perpendicular the measure was to the wall of the aorta. CT angiography (i.e., with iv contrast) provides additional information about intramural hematoma, thrombosis, and dissection. Detailed 3D reconstructions can be used to help operative planning. The abdominal aorta is always imaged as 25% of patients with thoracic aneurysms also have abdominal aneurysms. CT angiography can be performed to image coronary and carotid anatomy and identify lesions. In patients with renal insufficiency it is important to ensure adequate hydration and avoid nephrotoxic drugs in the days before and after a contrast CT as the dye is nephrotoxic.

MRI

MRI scanners are less widely available than CT scanners. They give accurate and reproducible information about size, anatomy, and extent of the disease in transverse, coronal and sagittal views, as well as dynamic images that can pinpoint intimal tears, assess flow in true and false lumens of dissected aneurysms, differentiate between periaortic hematoma and thrombosed false aneurysm, assess LV function, and peripheral vascular lesions. Contraindications to MRI include permanent and temporary pacemakers, ICDs, intracranial aneurysm and AVM clips, infusion pumps, cochlear implants, and metal fragments in the eye.

Aortography

Aortography provides accurate information of the location of the aneurysm in relation to major branches of the aorta, including the carotid arteries, as well as delineating patterns of blood flow within the aorta and any false lumen, and to viscera. Information about aortic regurgitation can be gained, as well as LV function. Coronary angiograms can be carried out, a useful adjunct in a population at risk of coronary artery disease. Annuloaortic ectasia appears as a pear-shaped dilatation of the aortic sinuses tapering to a narrower normal aorta. The main disadvantage of aortography is that it does not give any information beyond the lumen of the aorta, and may underestimate the size of aneurysms where there is mural thrombus. Aortography has only 84% sensitivity in diagnosing type B dissections. The

contrast used is nephrotoxic. Aortography is an invasive procedure that may be difficult to carry out in patients with severe peripheral vascular disease.

Other investigations

Cardiovascular system

- EKG with stress testing if indicated.
- TTE: mitral valve disease frequently accompanies aortic aneurysms, and in Marfan disease the incidence is as high as 30%.
- Coronary angiography for male patients >35 years and females >45 years, or with a history suggestive of ischemic heart disease: 25% of patients undergoing repair of ascending aneurysms have concomitant coronary artery disease.

Respiratory system

- CXR.
- Lung function tests in patients with COPD or long history of smoking.
- Spirometry and ABGs will assist the anesthesiology assessment of whether these patients will tolerate single lung ventilation, and establish a baseline function.
- Early admission for antibiotics, nebulized bronchodilators, and physiotherapy improves outcome in patients with significant COPD.

Renal function

- Urea and creatinine, GFR.
- Preoperative hydration, fluid balance monitoring, and occasionally admission to ICU for invasive monitoring and inotropic support in an attempt to optimize function may be indicated.

CNS function

Consider duplex of the carotid arteries in patients with severe atherosclerotic disease, or a history of TIA or stroke.

Management of aortic aneurysm

Natural history

Thoracic aortic aneurysms have a 5-year survival of about 40%, compared to 20% for thoraco-abdominal and abdominal aneurysms. The 5-year survival of chronic dissecting aneurysms may be as low as 13%. Thoracic aneurysms increase in size by about 0.4cm per year, with arch aneurysms increasing fastest. Larger aneurysms (>6cm) expand faster than smaller aneurysms, predicted by Laplace's law: aneurysms >8.0cm in diameter increase in size by up to 0.16cm per year. ***The size of aneurysm at presentation is the most important predictor of risk of rupture.*** The annual risk of rupture of aneurysms between 6.0–6.9cm is fourfold that of aneurysms of 4.0–4.9cm. Median size of rupture or dissection of ascending aneurysms is 5.9cm, and 6.9cm for descending aneurysms. Once symptoms develop the mean time to rupture is about 2 years. ***The annual risk of rupture, dissection, or death is 5–6% for aneurysms 5.0–5.9cm, and 10–15% for aneurysms >6.0cm. Elective operative mortality is:***

- 1–5% for elective repair of ascending aneurysms, with 85% 5-year survival (operative mortality of aortic root replacement is as low as 1% in experienced centres, but nationally mortality is closer to 7%).
- 5–10% for repair of arch aneurysms, with 75% 5-year survival.
- 5–10% for repair of descending aneurysms, with 60% 5-year survival.
- 5–10% for repair of thoracoabdominal aneurysms, with 60% 5-year survival.

Mode of death in unoperated thoracic aneurysms

- Rupture (40–70%).
- Dissection.
- Cerebral embolism.
- Visceral embolism.

Indications for surgery

Ascending aorta

- Acute dissection (emergency).
- Symptomatic aneurysms.
- Asymptomatic aneurysms >5.5cm.
- Aneurysms >5.0cm and known connective tissue disorder.
- Aneurysms >5.0cm and bicuspid aortic valve requiring replacement.
- Moderate to severe aortic regurgitation.

Aortic arch

- Symptomatic aneurysms.
- Aneurysms >5.5cm.
- Documented progressive enlargement.
- Potential source of emboli in patient with embolic cerebral events.

Descending aorta

- Symptomatic aneurysms.
- Asymptomatic aneurysms >6.5cm.

- Documented progressive enlargement.
- Aneurysms associated with chronic type B dissection.

Contraindications to surgery

- Advanced age (>85 years).
- Incurable malignancy or other terminal illness.
- Irreversible, profound brain injury, chronic dementia.

Controversies

Borderline aorta (4.5–4.9cm)

In light of newer data on rates of dissections in aorta <5.0cm, and low mortality in experienced centers with elective intervention, the 2008 AHA/ACC and 2007 ESC guidelines say essentially the same thing: repair/replacement of the ascending aorta and/or root is indicated for:

- Any patient if the diameter of the ascending aorta >5.0cm, or the rate of increase is >0.5cm/year.
- Patients undergoing aortic valve replacement if the ascending aorta >4.5cm.
- Lower thresholds may be considered in young patients of small stature.
- 2010 AHA/ACC thoracic aorta guidelines recommend cut-offs of 4.5cm for patients with Loeys–Dietz syndrome, or the TGFBR1 and 2 mutations.

Root replacement or interposition graft with aortic valve replacement

The decision whether to replace the coronary sinuses boils down to a balance between the incremental operative mortality of a complete root replacement compared to AVR + interposition graft (3–5%), and the long-term risk of dissection or aneurysm formation of the residual native tissue. In older patients without an obvious connective tissue disorder, an interposition graft is probably most reasonable as the risk of late complications from the native aortic sinuses is low and root replacement is associated with higher mortality. In younger patients with Marfan or other connective tissue disease, complete root replacement or valve-sparing root replacement is more frequently indicated as it can be performed with low operative mortality and they are otherwise at higher risk of late complications from the native aortic sinuses.

Indexing aortic size to body surface area

There are several methods of adjusting these cut-offs for patient size:

- Svensson *et al.*[1] reported that if aortas were replaced when πr^2 (cm)/(height (m) >10 that 95% of dissections in their retrospective series would have been avoided.
- Ergin *et al.*[2] recommended intervention based on multiples of the normal diameter for a 40-year-old patient with a BSA of $2m^2$, ranging from 1.3 in Marfan disease to 1.5 in asymptomatic patients with no history of connective tissue disorder.

References

1. Svensson LG *et al.* Relationship of aortic cross-sectional area to height ratio and the risk of aortic dissection in patients with bicuspid aortic valve *J Thorac Cardiovasc Surg* 2003; **126**:892–3.
2. Ergin MA *et al.* Surgical treatment of the dilated ascending aorta: when and how? *Ann Thorac Surg* 1999: **67**:1834–9.

Surgery for ascending aneurysms

Key facts

- The aim of surgery is to resect all aneurysmal aorta, anastomosing tube graft to relatively normal aorta. Depending on which segments of the proximal ascending aorta are affected this may involve a simple interpositional graft, replacement of the aortic valve, replacement or remodelling of the aortic root.
- If the aneurysm is quite proximal it may be possible to perform the distal anastomosis with the cross-clamp on. More distal involvement requires removal of the cross-clamp, to do a hemiarch replacement under 15–20min of hypothermic circulatory arrest (📖 p435).

Set-up

Sternotomy. Axillary or femoral (📖 p120) or low aortic cannulation (the cannulation site will be excised with the aneurysmal segment under DHCA). High aortic cannulation only if intention is to leave the distal ascending aorta in place, and perform anastomosis with clamp on.

Operative techniques

Interposition graft

This involves replacing the aneurysmal aorta with a tube graft. The aortic valve may or may not be replaced. The sinuses are preserved and so there is no need to reimplant the coronaries. It is usually possible to perform without circulatory arrest. The technique is described on 📖 p434.

- It is suitable for aneurysms limited to the ascending aorta and not involving the aortic root (proximal to the sinotubular junction).
- In young patients with dilated sinuses, or in patients where there is evidence of chronic dissection or other pathology (e.g., abscess formation) involving the sinuses or annulus root replacement is required.
- In elderly patients with moderate sinus dilatation an interposition graft is usually a better option than a root replacement as the risk of complications from leaving the sinuses intact is probably lower.
- In acute type A dissection it is preferable to perform an interpositon graft, even if there is some evidence of dissection into the sinuses, as root replacement in this setting is much more difficult (tissues are friable, bleeding may be insurmountable). Dissected sinuses can be reapproximated with bioglue, a Teflon strip, and a running 4/0 Prolene prior to performing the proximal interposition graft anastomosis.

Root replacement

This is indicated when ascending aortic aneurysms involve the aortic root, and severe valvular disease means that it is not possible to preserve the aortic valve. The aortic root and aneurysmal aorta is replaced with a composite conduit with a mechanical (Bentall) or bioprosthetic valve (bio-Bentall) or a homograft aortic root and aorta harvested *en bloc* from a cadaveric human heart. This requires mobilization and reimplantation of the coronary ostia. The technique is described on 📖 pp386–391.

Ross procedure

This technique involves aortic root replacement with the patient's own pulmonary trunk, which is in turn replaced with a pulmonary homograft, and more limited indications. The technique is discussed on 📖 pp388–389.

Aortic valve-sparing procedures

Several techniques reduce the size of the aortic root without replacing the aortic valve, although it is usually resuspended within the tube graft, which replaces the resected segment of aorta. These techniques are described on 📖 p446. External wrapping of the aorta is very occasionally used in patients too frail to undergo a more extensive procedure.

Tips and pitfalls in aortic root surgery

Cabrol graft

This is useful in aortic root replacement where the coronary buttons are too friable, adhesed, calcified, or fibrosed to permit mobilization enough for them to be anastomosed to aortic conduit without tension. It is a particularly useful adjunct in reoperative root replacement, if the original aortic conduit has to be excised. A length of SVG, or more commonly an 8mm Dacron graft, is sewn end-to-end to the left coronary button with continuous 6/0 Prolene (buttressed if necessary with a 2mm Teflon felt strip) wrapped round the graft so that it can be sewn end-to-side to the right coronary button, and then end-to-side to the aortic conduit. In *hemi-Cabrol* just one coronary button is anastomosed in this way.

Cabrol patch or shunt

This is a bail-out technique for uncontrollable hemorrhage at the end of the case, usually from the proximal anastomotic or coronary button suture lines. In the original description the RA appendage was sewn to the aneurysmal sac, decompressing the perigraft space and allowing autotransfusion to the patient. Numerous modifications exist. Sewing a 10 x 8cm bovine pericardial patch to all the anterior surface of the heart and great vessels (usually the SVC, RA, RVOT, PA, ascending aorta), and just prior to closure creating a 1–2cm incision in the RA underneath the patch, works well to autotransfuse all blood loss into the RA, if there is no connection between the aortic root and the posterior pericardial sac. If the root communicates with the posterior pericardium, packing the posterior pericardial well and waiting before chest closure usually works surprisingly well. If the patch appears tense, you need to enlarge the right atrial hole (can be done via the suture line or a purse-string). These patches can be left in place (the shunt eventually disappears) or taken down a day or so later—remarkably little bleeding from the right atriotomy is seen at takedown in most patients.

Right coronary ischemia

The right coronary button is more frequently the source of problems that can compromise separation from bypass, than the left button—possibly because it is more prone to distortion and kinking as the heart fills. Even a tiny non-dominant RCA can result in severe RV dysfunction if it is not accurately reimplanted: have a low threshold for performing a hemi-Cabrol or a bypass graft.

Valve-sparing surgery

Key facts

- Important aortic root pathology necessitating surgical correction frequently coexists with ***structurally normal aortic valve leaflets***.
- Any technique that allows root reconstruction without valve replacement avoids some of the disadvantages of aortic valve replacement (📖 pp370–371). A large number of valve sparing root techniques have evolved, which can be divided into two main groups:
  - ***Remodeling*** techniques, e.g., the ***Yacoub*** in which a tailored graft is sewn to the remnants of the individual aortic sinuses, preserving dispensability and creating neosinuses, but not reinforcing the annulus, in a technically slightly easier procedure (Fig. 9.6).
  - ***Reimplantation*** techniques, e.g., the ***David*** in which the whole aortic root is replaced by a Dacron tube, secured to the annulus, in a technically more demanding procedure (Fig. 9.6)

Valve-sparing techniques

Remodeling: 1979 Yacoub

A Dacron graft, sized to the internal aortic diameter, is tailored to individually replace the resected sinuses (Fig. 9.6). The commissural pillars are resuspended within the Dacron graft so that the leaves of the graft replace each coronary sinus, and so that they are stretched upwards slightly maintaining tension on the valve leaflets. The coronaries are reimplanted as buttons.

- The technique is supposed to be technically easier than the David.
- Disadvantages are that the annulus is not stabilized (potentially a problem in patients with connective tissue disorders), and the sinotubular junction is not narrowed, so connective tissue disorders are a relative contraindication (long-term risk of annular dilatation and recurrent AI).
- Most useful in smaller roots because of the more physiologic dynamics.

Reimplantation: 1992 Tirone David I (TDI), 1995 TDII, Seattle, 1996 TD III, 1997 TD IV, 2002 TD V

This technique was designed for patients with dilatation of the sinotubular junction, dilation of the sinuses of Valsalva and annulo-aortic ectasia. The aortic valve is reimplanted within a Dacron tube-graft (Fig. 9.6) sized to the external aortic diameter. The root is excised leaving the ostia on 10mm buttons of sinus aorta, and the proximal coronary arteries mobilized. The Dacron graft is parachuted down onto the base of the aortic root, via pledgetted sutures placed through the base of the aortic outflow tract. The valve commissures are resuspended within the Dacron graft, the coronary buttons sewn into holes created within the tube graft, and the distal anastomosis performed. The annulus is stabilized by the Dacron graft. The TD III added a Teflon felt strip to buttress the annulus. The 1995 Seattle and the TD IV and V scalloped and plicated larger grafts to create more bulbous neosinuses.

- The technique has a longer learning curve than the Yacoub.
- The advantage is that the annulus is stabilized, so patients with annuloaortic ectasia and connective tissue disorders may be less prone to late annular dilatation and recurrent AI.
- Early iterations did not recreate sinuses, changing the root dynamics, contributing to abnormal leaflet stresses and late recurrence of AI.

Choice of technique

Lesion	Causes	Technique
Annulus normal, STJ enlarged	Aortic ectasia Aneurysm	Yacoub
Annulus and STJ normal	Aortic dissection	Interposition graft and glue sinuses, or replace non-coronary, or Yacoub
Annulus and STJ enlarged	Annuloaortic ectasia Marfan	David, or interposition graft

Patient outcomes

Perioperative elective mortality is approximately 1–2% (young patients):

- 10-year survival is 80–90% (close to normal population).
- Freedom from reoperation at 10 years is highly variable—while Tirone David reports 90–98%, this is lower elsewhere (75% where the Yacoub was used in Marfan syndrome), and not a direct reflection of freedom from mod-severe AI which was 85% at 10 years in the David series, i.e., not dissimilar to bioprosthetic AVR.

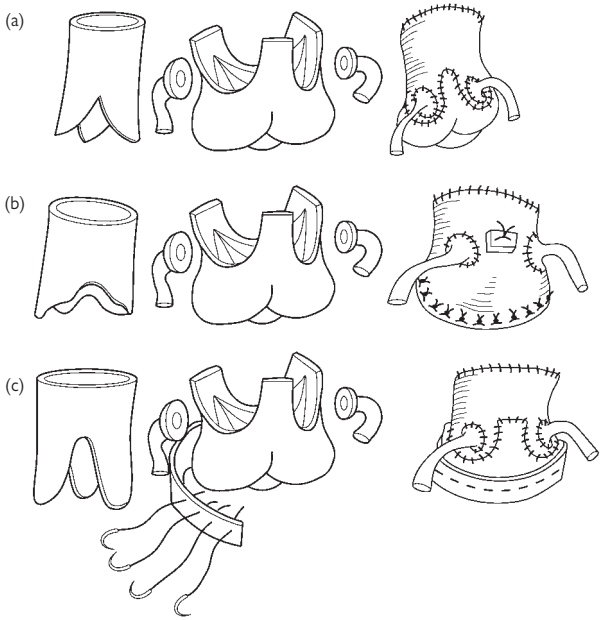


Fig. 9.6 Examples of valve sparing aortic surgery. (a) Yacoub repair. (b) Seattle repair. (c) Tirone III repair.

Arch replacement

Key facts

- Aneurysms and type A dissections may involve the distal ascending aorta and arch: the main options for arch replacement are:
  - ***Island*** technique (Fig. 9.7b).
  - ***Trifurcated graft*** (Fig. 9.7d) which has largely superseded the Island technique as it is easier to selectively perfuse the brain reducing DHCA, replaces the entire arch, and anastomosis is limited to the usually less diseased head and neck vessels.
  - ***Stage I elephant trunk*** (Fig. 9.7a) is an adjunct to either of the above techniques where the aneurysmal descending aorta will need later replacement (usually >3 months after the stage I).
- Aortic arch surgery mandates cerebral protection in addition to myocardial protection (📖 p124). DHCA is almost invariably required as the distal anastomosis must be performed with the cross-clamp off and the head and neck vessels must be reimplanted if the arch is replaced.
- DHCA may be supplemented by direct antegrade perfusion, selective cerebral perfusion, or retrograde cerebral perfusion (📖 p124).

Set-up

- Set-up is described on 📖 p444: an extended sternotomy may be used so prep and drape to expose neck and axillary cannulation sites.
- Bypass is instituted via fem-bicaval, axillo-caval, or fem-fem cannulation (which can be changed to aorto-caval cannulation at a later point) and the patient is cooled to 15°C.
- Cardioplegia is given retrograde and/or antegrade, and a right superior pulmonary vein vent usually placed.

Operative technique

- When the nasopharyngeal temperature is 15°C, the patient is placed in Trendelenburg, caval tapes secured, and the SVC pipe clamped temporarily to eliminate negative pressure in the cerebral venous system that might entrain air emboli when the aorta is opened.
- Dissect the phrenic and the vagus nerves from the anterior surface of the aorta, and dissect out the proximal 1cm of the innominate artery, left common carotid, and subclavian.
- Select and trim a trifurcation graft to fit: sizes of the two distal arms are usually 8mm or 10mm, and the graft may have an additional side branch for later aortic cannulation.
- The pump is stopped, the cross-clamp is removed, and blood vented from the arch so that the arch and vessels can be inspected.
  - If the dissection flap or aneurysmal portion does not extend into the arch an anastomosis with the distal ascending aorta can be carried out as described on 📖 p435.
  - If the dissection flap or aneurysmal portion extends into the arch, but does not extend into the descending aorta or arch vessels, the aorta is transected obliquely and an anastomosis fashioned similarly under 15–20min DHCA: this is a ***hemiarch replacement***.
  - If the pathology involves the whole arch, then it must be replaced, and the arch vessels reimplanted: the *trifurcated graft technique* is described next.

- Transect the aorta distal to the origin of the left subclavian, and the head and neck vessels close to their origin (Fig. 9.7).
- Using continuous 4/0 Prolene on an SH needle starting forehand well to the left of the heel, inside to out on the Dacron and outside to in on the native vessel, running the suture round to the heel and then up side nearest you, then taking the other arm and running it forehand towards you across the top: complete the left subclavian, then left common carotid and then innominate artery anastomosis. The Dacron can be bevelled to fit.
- If axillary cannulation was used, the pump is then resumed at 50mL/min to de-air the graft, the proximal end of which is then clamped and selective cerebral perfusion commenced (📖 p125).
- A Dacron graft is selected to fit the distal anastomosis (24–30mm) and using a strip of Teflon felt to buttress either sewn end-to end or as an elephant trunk (where the graft is invaginated with the inner portion then pulled out to complete the proximal anastomosis (Fig 9.7a), while the outer portion marked with three metal clips remains in the descending aorta for the Stage II descending repair). This can be de-aired and clamped and perfusion begun to the lower body.
- The proximal ascending aorta anastomosis, which may involve a root replacement (📖 p444) or simple interposition graft (📖 p434) is completed, and a graft-to-graft anastomosis using 4/0 Prolene between the proximal and distal grafts is performed (taking care to leave a landing zone for the trifurcated graft), and normal cardiopulmonary bypass resumed.
- The trifurcated graft is anastomosed finally, selective cerebral perfusion is over and the patient is rewarmed.

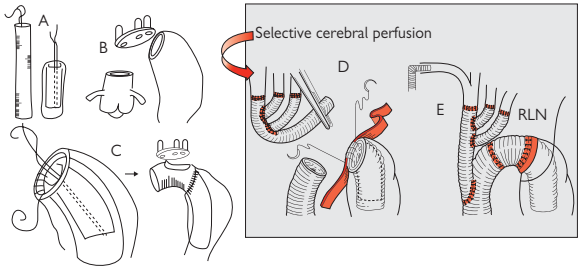


Fig. 9.7 Arch replacement. (a) Invaginating Dacron graft for elephant trunk. (b) Set-up for Island technique. (c) Stage I elephant trunk using island technique for arch. Gray box shows trifurcated graft technique. (d) Stage I elephant trunk with selective cerebral perfusion. (e) Completed arch replacement with trifurcated graft.

Repair of descending aneurysms

Overview

The natural history and pathology, are discussed on 📖 p48. The aim of surgery is to resect all aneurysmal aorta, reimplanting the larger radicular arteries and all involved visceral arteries. If the arch is not involved circulatory arrest is not required, and it may be possible to use partial left heart bypass instead of full cardiopulmonary bypass (📖 p125). The variable extent of thoracoabdominal aneurysms makes accurate preoperative imaging vital. Spinal protection is an important part of surgery on the descending aorta as the blood supply to the spinal cord is usually compromised (📖 p436) resulting in paraplegia rates as high as 20%. Techniques of spinal protection are described on 📖 p437, including:

- Surgical technique.
- Spinal fluid withdrawal.
- Hypothermia and topical cooling.
- Pharmacological adjuncts.

Set-up

This is as described on 📖 p438. Double-lumen ET tube for single-lung ventilation. The patient must be positioned in the right lateral position with the hips rotated so that the left groin is accessible.

Incision

Crawford type I and II aneurysms are approached via the 6th rib space through a lateral thoracotomy extended via a gentle curve into a midline laparotomy incision extending to the umbilicus if required (Fig. 9.5). For Crawford type III and IV aneurysms the incision is through the 8th or 9th rib spaces.

- Deflate the lung.
- The diaphragm is divided leaving a 1–1.5cm rim of tissue at the thoracic wall, to avoid damage to the phrenic nerves.
- Divide the retroperitoneum lateral to the descending colon.
- Develop the plane anterior to psoas, posterior to the left kidney as far as the lateral aspect of the abdominal aorta.
- Retract the descending colon, spleen, tail of pancreas, and left kidney to the right.
- Divide the crux of the diaphragm to expose the celiac axis, renal and mesenteric vessels.
- Although the entire approach can be retroperitoneal particularly where extensive adhesions from previous laparotomies make an open approach difficult, an open approach makes it possible to asses the viscera after reperfusion.

Venting

The heart can be vented through the left inferior pulmonary vein, or the apex.

Options for cardiopulmonary bypass

'Clamp and go'

Cardiopulmonary bypass is not used, the patient is not cooled. There is no control of upper body hypertension on clamping the descending aorta, and no perfusion to the lower body during this time. It is used in emergencies if cardiopulmonary bypass is not available, and in elective situations may be preferred by some surgeons.

Full cardiopulmonary bypass with deep hypothermic circulatory arrest

The technique of deep hypothermic circulatory arrest is described on 📖 p124. In repair of descending aortic aneurysms the following options for cannulation are available:

- Fem-fem cannulation.
- Fem-fem bypass augmented with flow from the PA.
- Axillo-femoral bypass.
- Arterial drainage from left subclavian or distal aorta.

CPB is instituted as soon as possible and cooling started. Various methods of spinal cord protection may be used (📖 p437). The heart is vented as soon as it starts to fibrillate. Once the nasopharyngeal temperature is 15°C venting is stopped (to prevent entraining air into the heart) and bypass is discontinued. The aorta distal to the aneurysm is either clamped or, if this is not possible due to heavily calcified atheroma or poor access, a Foley catheter balloon is inflated with saline to occlude the lumen.

Partial left heart bypass

Left heart bypass consists of drainage of oxygenated blood via a cannula placed in the LA or one of the left pulmonary veins, which is then pumped using an in line centrifugal pump into either the left common femoral artery or the distal aorta. Bypass flows of up to 1.5L/min are adjusted to maintain a distal arterial pressure of 60–70mmHg. The upper body circulation proximal to the cross-clamp maintains near normal perfusion pressures. The aorta is clamped either side of the aneurysmal segment. Depending on the extent of the aneurysmal segment this allows constant perfusion of abdominal visceral, lower limbs, and spinal cord. As each arterial branch is reanastomosed to the graft the clamp can be moved distally. The advantages of left heart bypass are:

- Reduced risk of stroke from aortic cannulation.
- Reduced risk of aortic dissection.
- Reduced pulmonary dysfunction postoperatively.
- Some surgeons use a reduced dose of heparin (1000U/kg).

Traumatic aortic transection

Key facts

Traumatic aortic transection is caused by massive deceleration injuries. The force needed to rupture the aorta is equivalent to an intraluminal pressure of >2000mmHg. High-speed impacts result in continued forward movement of the distal end of the aortic arch, against the upper descending aorta which is held stationary relative to the thorax by the ligamentum arteriosum, intercostal arteries, and origins of the head and neck vessels. As a result almost 50% of acute aortic transactions occur in the very proximal descending aorta near the aortic isthmus, 1cm distal to the origin of the left subclavian. <10% occur in the ascending aorta. The remainder occur in the distal thoracic aorta, often in combination with vertebral fractures.

- Transection may be complete, including adventitia and pleura.
- Adventitia and/or pleura may be spared.
- The aortic intima may be spared.
- False aneurysms result from partial transactions: they calcify, enlarge, and eventually often rupture.

Natural history

When transection is complete death is almost instantaneous as there are no tissues to tamponade hemorrhage. Tears that spare the adventitia or pleura result in proportionately smaller hemorrhages into the retroperitoneal tissues that result in severe chest pain radiating through to the back, hemorrhagic shock, and left hemothoraces. Risk of death is greatest immediately after the transection, and although the instantaneous risk of death decreases with time it levels off after 7 days leaving a low constant risk of death from hemorrhage.

- 48% of untreated patients are dead within 48h.
- At 10 years the predicted 5-year mortality from rupture is 20%.

Diagnosis

The classical features of a widened mediastinum on CXR may not be apparent: features to look for include rightward shift of the trachea, blurring of the sharp outline of the proximal descending aorta, and opacification of the aorto-pulmonary window. Aortography is being replaced by MRI as the gold standard for imaging, although the usual caveats about availability of MRI apply. Spiral CT has a lower specificity, but may be vital in the diagnosis and management of associated injuries.

Indications for surgery

Urgent surgical repair is indicated for stable patients with contained aortic injury who do not require laparotomy, craniotomy, pelvic stabilization, or thoractomy for bleeding. Emergency repair is indicated for:

- Signs of impending rupture.
- Hemothorax.
- Contained hematoma.

After 10–14 days aortic transactions may be treated as chronic aortic aneurysms (📖 p450). Hypertension should be avoided.

Outcome

Variability in outcome reflects the range in severity of transection ranging from moribund patients exsanguinating from complete tears, to stable patients with contained hematomas.

- The operative mortality in the former group is >40%.
- This compares to <5% in the latter group.
- Overall hospital mortality is 30%.

Technique

The techniques are similar to those described on 📖 pp438–439. The same adjuncts for spinal cord protection are often not feasible in the critically ill trauma patient. The main differences between the technique described on 📖 pp438–439 for repair of a type B dissection, and repair of a traumatic transaction are:

- Dissection around the isthmus (the usual location of a tear) is only started once proximal and distal aortic control has been established.
- 'Clamp and go', where the aorta is simply cross-clamped just distal to the left subclavian artery, and the section including the tear is replaced with a Dacron graft, is useful in trauma centers where extracorporeal bypass may not be an option, and for rapid control of continued hemorrhage: cross-clamp times of <30min result in low paraplegia rates, but most cross-clamp times are significantly longer.
- Lower body perfusion systems, e.g., left heart partial bypass (📖 p451) (if available) should be chosen with the aim of using the least amount of heparin possible, because of the risk of concomitant intracranial hemorrhage.

Endoluminal stenting

- Endoluminal stenting is increasingly used instead of surgery for traumatic aortic rupture, as it appears to offer similar short-term survival, without the need for thoracotomy, single lung ventilation, or cardiopulmonary bypass.
- Gore Tag® grafts are used to cover relatively short segments (5–10cm).
- There is little long-term outcome data available.

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